APOLLO RESEARCH

Research Scientist (Control)

APPLY FOR THIS JOB

London & San Francisco
Product / Full-Time / On-Site

THE OPPORTUNITY

Join our new AGI safety product team and help transform AI control research into practical tools that directly reduce risks from AI. As an **Research Scientist (Control)**, you'll work closely with Marius (CEO & currently leads the monitoring efforts), other control researchers and product engineers.

We are currently building Watcher, a monitoring tool for coding agents. Our monitoring research agenda attempts to translate compute into safety at scale. You will join a small team and will have significant ability to shape the team & tech, and have the ability to earn responsibility quickly.

You will like this opportunity if you're passionate about using empirical research to make AI systems safer in practice. You enjoy the challenge of translating theoretical AI risks into concrete detection mechanisms. You thrive on rapid iteration and learning from data. You want your research to directly impact real-world AI safety.

KEY RESPONSIBILITIES

TLDR: you will design & implement control protocols (see e.g. [Greenblatt et al, 2023]) and test them on real-world production systems at scale.

Research & Development

- Design and conduct experiments to test monitor effectiveness across different failure modes and agent behaviors
- Build and maintain evaluation frameworks to measure progress on monitoring capabilities
- Build and maintain high-quality datasets to train and test monitors on
- Iterate on monitoring approaches based on empirical results, balancing detection accuracy with computational efficiency
- Contribute newly discovered coding agent failure modes to Apollo's shared failure mode library (owned by the AI Security & Control Researcher/Engineer) as they surface during monitor design, evaluation, and iteration
- Stay current with research on AI safety, agent failures, and detection methodologies
- Stay current with research into coding security and safety vulnerabilities

Fine-tuning & Monitor Development

- Fine-tune open-source models to create efficient monitors for high-volume production environments
- Design and build agentic monitoring systems that autonomously investigate logs to identify both known and novel failure modes
- Incorporate adversarial findings from red-teaming campaigns into monitor design and evaluation, closing the loop between attack and defense

JOB REQUIREMENTS

Must-haves

- 2+ years of experience conducting empirical research with large language models or AI systems
- Strong experience with AI coding agents. For example, having extensively used and compared frontier coding agents, or having designed / developed coding agents
- Experience with LLM-as-a-judge setups or AI monitoring more broadly
- Experience designing and running experiments, analyzing results, and iterating based on empirical findings e.g. prompting, scaffolding, agent design, fine-tuning, or RL
- Strong Python programming skills
- Demonstrated ability to work independently on open-ended research problems

Good-to-haves

- Familiarity with AI safety concepts, particularly agent-related risks
- Familiarity with computer security, e.g. security testing and secure system design
- Experience fine-tuning language models or working with smaller open-source models
- Previous work building developer tools or monitoring systems
- Publications or contributions to AI safety or ML research
- Experience with production log systems or production log analysis

We want to emphasize that people who feel they don't fulfill all of these characteristics but think they would be a good fit for the position nonetheless are strongly encouraged to apply.

REPRESENTATIVE PROJECTS

- **Hierarchical monitoring for coding agent security**: Design a multi-layer monitoring system for detecting security vulnerabilities introduced by coding agents. Start by cataloging common security failure modes (e.g., hardcoded credentials, SQL injection vulnerabilities, insecure API calls). Build specialized monitors for each category, then create a hierarchical system where fast, efficient first-pass monitors flag potentially problematic code for deeper investigation by more sophisticated monitors.
- **Design a backtesting strategy for Watcher:** Validate the system on synthetic test cases and real agent outputs, iterating to optimize the tradeoff between detection rates and false positives while maintaining low latency for most monitoring decisions.
- **Fine-tune an open-source model to be a great monitor:** Take an open-source model and fine-tune it on our large dataset of coding agent failures with high-quality ground truth labels. Test different fine-tuning techniques and measure generalization to our held-out data. Compare against prompted baselines on accuracy, cost and latency. The goal is to fine-tune models to lift the pareto frontier of monitors.

BENEFITS

- **This role offers market competitive salary, equity, and competitive benefits.**
- Salary: **San Francisco**: $204,000 – $385,000; **London**: £136,000 – £258,000. We will be looking to meaningfully raise salaries soon.
- Flexible work hours and schedule
- Unlimited vacation
- Unlimited sick leave
- Up to 6 months of paid parental leave
- Comprehensive health, dental and vision insurance
- Retirement savings with competitive employer matching (e.g. 401(k) for US employees)
- Lunch, dinner, and snacks are provided for all employees on workdays
- Paid work trips, including staff retreats, business trips, and relevant conferences
- A yearly $1,000 (USD) professional development budget
- Relocation support and visa fees (if applicable)

LOGISTICS

- **Time Allocation**: Full-time
- **Location**: This is an in-person role working out of our London or San Francisco office. We offer flexible working hours and wfh arrangements.
- **Visa sponsorship:** We sponsor visas in both the UK and US. Sponsorship isn't guaranteed for every role or candidate, but if we make you an offer, we'll work with you to find the right visa route.

ABOUT THE TEAM

The product team consists of research scientists: Víctor Gillioz, Monika Jotautaitė, Dmitrii Volkov; product engineers: Jeremy Neiman, Zak Walters, Zen van Riel, Srdjan Miletic and Gustavo Bicalho; and our GTM lead: Kyle Daï. Marius Hobbhahn (CEO) advises the team. Furthermore you will interact with our other SWEs and researchers, since we intend to be "our own customer" by using our products internally for our research work. You can find our full team here.

ABOUT APOLLO RESEARCH

The rapid rise in AI capabilities offers tremendous opportunities, but also presents significant risks. At Apollo Research, we're primarily concerned with risks from Loss of Control, i.e. risks coming from the model itself rather than e.g. humans misusing the AI. We're particularly concerned with deceptive alignment / scheming, a phenomenon where a model appears to be aligned but is, in fact, misaligned and capable of evading human oversight.

We work on the science of scheming, detection of scheming (e.g. building evaluations), and scheming mitigations (e.g. anti-scheming). We also work on control and monitoring research (see our scalable monitoring agenda). We work closely with many frontier AI companies, such as OpenAI, Anthropic, Google, Meta, Thinking Machines and others, e.g. to test their models and collaborate on the science of scheming. At Apollo, we aim for a culture that emphasizes truth-seeking, being goal-oriented, giving and receiving constructive feedback, and being friendly and helpful. If you're interested in more details about what it's like working at Apollo, you can find more information here.

We also build a coding agent security product called Watcher that secures agent deployments in companies. Our goal is to reduce the probability of catastrophic incidents by securing coding agents, learning about their real-world risks, and publishing our research on how to build these control systems most effectively.

Equality Statement: Apollo Research is an Equal Opportunity Employer. We value diversity and are committed to providing equal opportunities to all, regardless of age, disability, gender reassignment, marriage and civil partnership, pregnancy and maternity, race, religion or belief, sex, or sexual orientation.

HOW TO APPLY

Please complete the application form with your CV. The provision of a cover letter is neither required nor encouraged. Please also feel free to share links to relevant work samples.

About the interview process: Our multi-stage process includes a screening interview, a take-home test (3 hours), 3 technical interviews, and a final interview with Marius (CEO). There are no leetcode-style general coding interviews. You may use AI tools on the take-home; we judge the result the way we'd judge any contributor's work, so you are responsible for the quality of everything you submit. If you want to prepare, we suggest building simple monitors for coding agents and running them on your own Claude Code / Cursor / Codex / etc. traffic.

Your Privacy and Fairness in Our Recruitment Process: We are committed to protecting your data, ensuring fairness, and adhering to workplace fairness principles in our recruitment process. To enhance hiring efficiency, we use AI-powered tools to assist with tasks such as resume screening. These tools are designed and deployed in compliance with internationally recognized AI governance frameworks. Your personal data is handled securely and transparently. All resumes are screened by a human and final hiring decisions are made by our team. If you have questions about how your data is processed or wish to report concerns about fairness, please contact us at info@apolloresearch.ai.

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